ALICE L WALTON SCHOOL OF MEDICINE

Drugs to Treat Hepatitis B and C Infections

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Adapted from: Drugs to Treat Hepatitis B and C Infections

Listen to this Brick
0:00 / 22:42 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Compare and contrast the mechanisms of action of agents used in the treatment of hepatitis B infections.
2. Compare and contrast agents the mechanism of action of agents used in the t
3. Discuss the adverse drug reactions of drugs used to treat hepatitis B and C infections.

CASE CONNECTION

"I was honestly shocked when they told me I had hepatitis C," says AR, a 27-year-old woman who presents to clinic after abnormal screening labs obtained during a routine physical exam. She reports a history of intravenous drug use, which she discontinued one year ago. She denies jaundice but describes intermittent fatigue and difficulty concentrating, which she attributed to stress at work. Laboratory testing confirms chronic hepatitis C infection, with detectable HCV RNA and mildly elevated transaminases. An abdominal ultrasound shows no evidence of cirrhosis. AR is concerned about long-term complications and asks, "I'm still young—what kind of treatment would I need? Are there medications that can actually get rid of this virus?" As you review her case, you consider the goals of hepatitis C therapy and how direct-acting antivirals target specific stages of the viral life cycle.

Based on her history, lab results, and absence of advanced liver disease, which antiviral agent would be most appropriate, and how does it work?

GO TO CONCLUSION ↓

What Are Drugs to Treat Hepatitis B and C Infections?

Hepatitis B is transmitted through sexual contact and from mother to child during vaginal birth, through blood exposure, including transfusions and intravenous drug use. Both viruses infect the liver and may cause symptoms such as jaundice, fatigue, and joint pain. At times, patients remain asymptomatic.

Without treatment, chronic infection can lead to cirrhosis, liver cancer, and liver failure. Treatment is therefore essential: while hepatitis B is not yet curable, many individuals clear the infection spontaneously, and effective therapies exist to control disease progression. Notably, newer antiviral drugs can cure hepatitis C.

Antiviral drugs are used against infection with viral hepatitis. As with other antiviral medications, they target viral metabolic products such as proteins, enzymes, and nucleic acids because viruses use the host organelles to replicate. Targeting viral products slows virus replication and spread while minimizing its effect on the host. These drugs are designed for specific proteins and metabolites of each virus and therefore have a narrow range of use.

Antiviral Drug Mechanisms of Action

Since most antivirals have a narrow range of use, medications effective against one viral hepatitis are not effective against the other. Hepatitis B is treated with nucleoside analogs that inhibit hepatitis B viral DNA polymerase. Hepatitis C treatment has recently changed from interferon α-2a therapy to direct-acting agents (DAAs). These virus-specific drugs act as protease inhibitors, RNA polymerase inhibitors, and viral assembly inhibitors. In both infections, antivirals are started if the patient is severely ill or fails to independently clear the infection.

Why are treatments for hepatitis B and C different?

How Do We Treat Hepatitis B Virus? (LO#1 and 3)

Acute infection is managed with supportive treatment and monitoring for resolution. Chronic HBV is treated with nucleoside and nucleotide analogs. These drugs prevent viral replication by acting on the viral DNA polymerase. HBV DNA polymerase has some reverse transcriptase activity, so some HIV medications are also effective in treating HBV.

CLINICAL CORRELATION

Most hepatitis B virus (HBV) infections are self-limited. Less than 5% of immunocompetent patients go on to have chronic HBV infection.

What is the mechanism of action of antiviral medications for chronic HBV?

Tenofovir is an acyclic analog of adenosine monophosphate. After entering cells, it is phosphorylated to tenofovir diphosphate, the active metabolite. Tenofovir diphosphate competitively inhibits viral reverse transcriptase by competing with natural dATP. When incorporated into viral DNA, it lacks a 3′-OH group, causing obligate chain termination. This prevents viral DNA synthesis, thereby blocking HIV replication (and HBV DNA polymerase activity) (Figure 1).

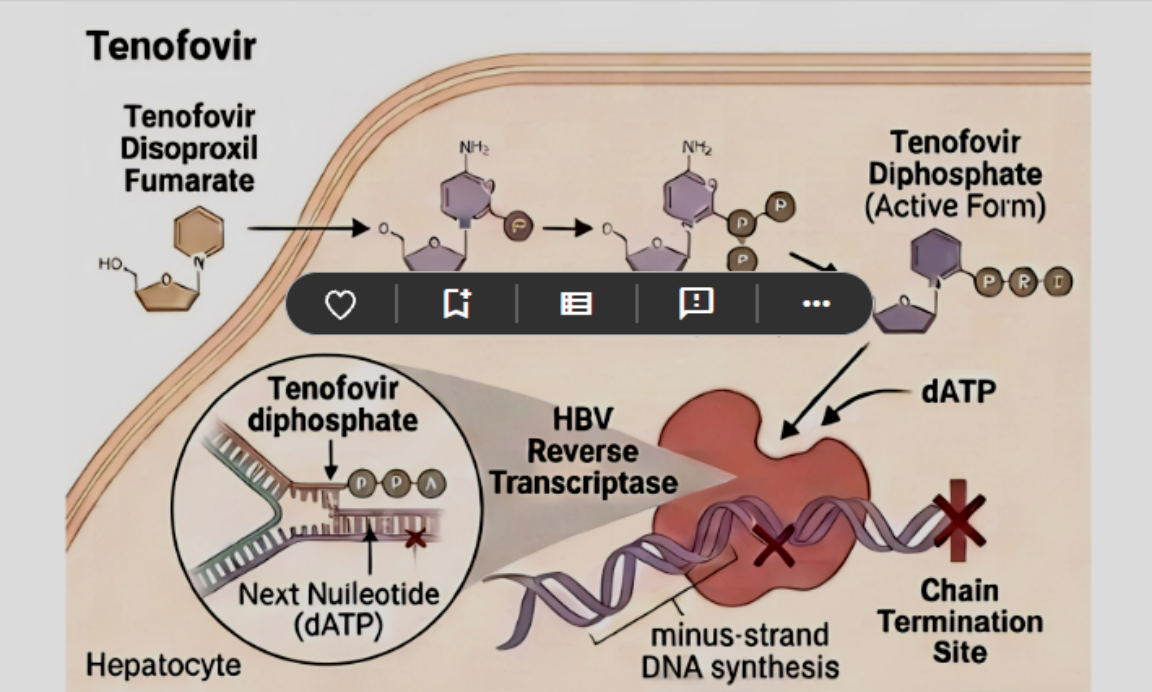

Tenofovir
Tenofovir Disoproxil Fumarate
Tenofovir Diphosphate (Active Form)
Tenofovir diphosphate
HBV Reverse Transcriptase
dATP
Next Nuileotide (dATP)
minus-strand DNA synthesis
Chain Termination Site
Hepatocyte

Figure 1: Tenofovir Mechanism of Action

Entecavir is a guanosine nucleoside analog for the treatment of HBV infection. Following intracellular phosphorylation to the triphosphate, it competes with the natural substrate, deoxyguanosine triphosphate, thereby inhibiting function of HBV polymerase. Thus, entecavir will:

1. Inhibit base priming
2. Inhibit reverse transcription of negative strand from pregenomic mRNA (pgRNA)
3. Inhibit the synthesis of the positive strand of HBV DNA (Figure 2)

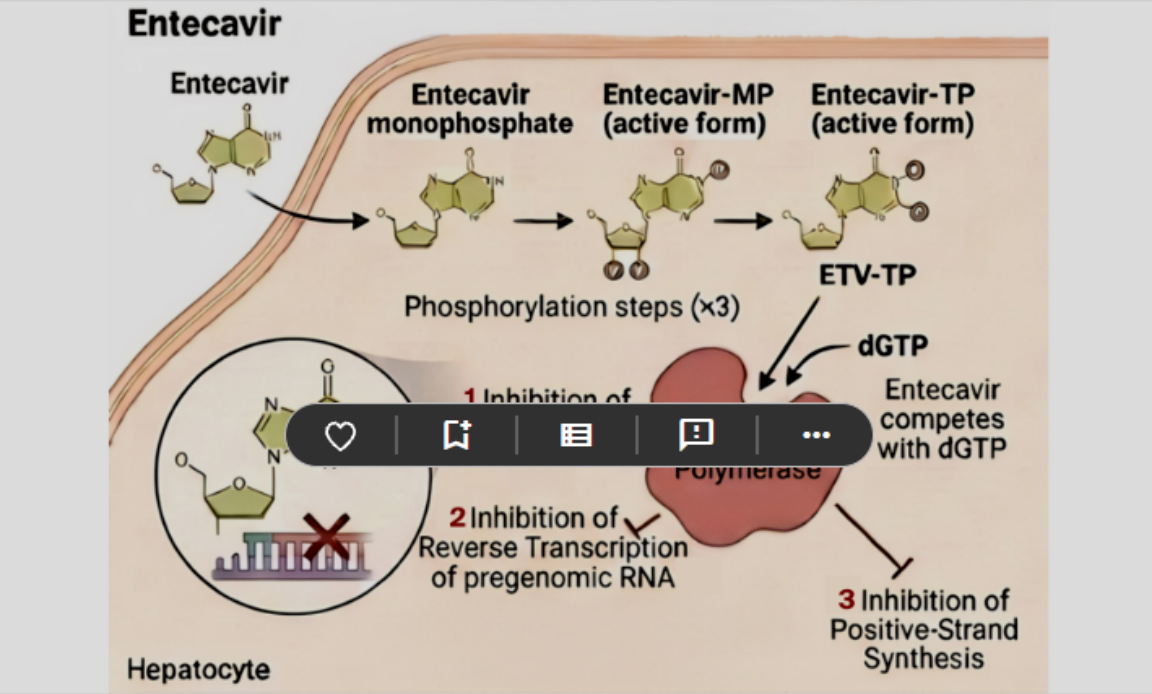

Entecavir
Entecavir
Entecavir monophosphate
Entecavir-MP (active form)
Entecavir-TP (active form)
Phosphorylation steps (×3)
ETV-TP
dGTP
Entecavir competes with dGTP
1 Inhibition of
Polymerase
2 Inhibition of Reverse Transcription of pregenomic RNA
3 Inhibition of Positive-Strand Synthesis
Hepatocyte

Figure 2: Entecavir Mechanism of Action

Entecavir is effective against lamivudine-resistant strains of HBV. common adverse effects include edema (16%), headache, and fatigue. Lactic acidosis and hepatotoxicity have been reported when used with other antiretroviral agents. Concomitant use of drugs with renal toxicity should be avoided.

A single nucleoside analog is effective in suppressing viral DNA count and elimination of hepatitis B e antigen, but the downside is the long treatment course.

The use of agents such as adefovir and lamivudine used to be common; however, they are not used due to the development of resistance.

Nucleoside therapy is a minimum of 4 years and may be indefinite depending on the patient's response. During the treatment course, the patient is regularly monitored for viral response and adverse effects.

Interferon α–2a

Hepatitis B may also be treated with interferon α-2a. Interferon α-2a activates genes and proteins that interfere with the viral life cycle and stimulates cell-mediated immunity to kill infected cells. Pegylated (containing a polyethylene glycol molecule) interferon α-2a is preferred over standard interferon α-2a because the dosing is

Pegylated interferon α-2a is administered subcutaneously once a week. Adverse reactions to the drug make it a difficult choice for patients because interferon α-2a has more side effects than nucleoside analogs. Approximately 19–28% of patients experience myalgias, arthralgia, insomnia, irritability, neutropenia, and transaminitis. Interferon should not be administered to pregnant patients.

It is generally reserved for patients who are immunocompetent, without cirrhosis, and want a shorter duration of treatment. Interferon α-2a therapy lasts 48 weeks and requires regular monitoring for adverse effects and resolution.

How Do We Treat Hepatitis C Infection? (LO#2 and 3)

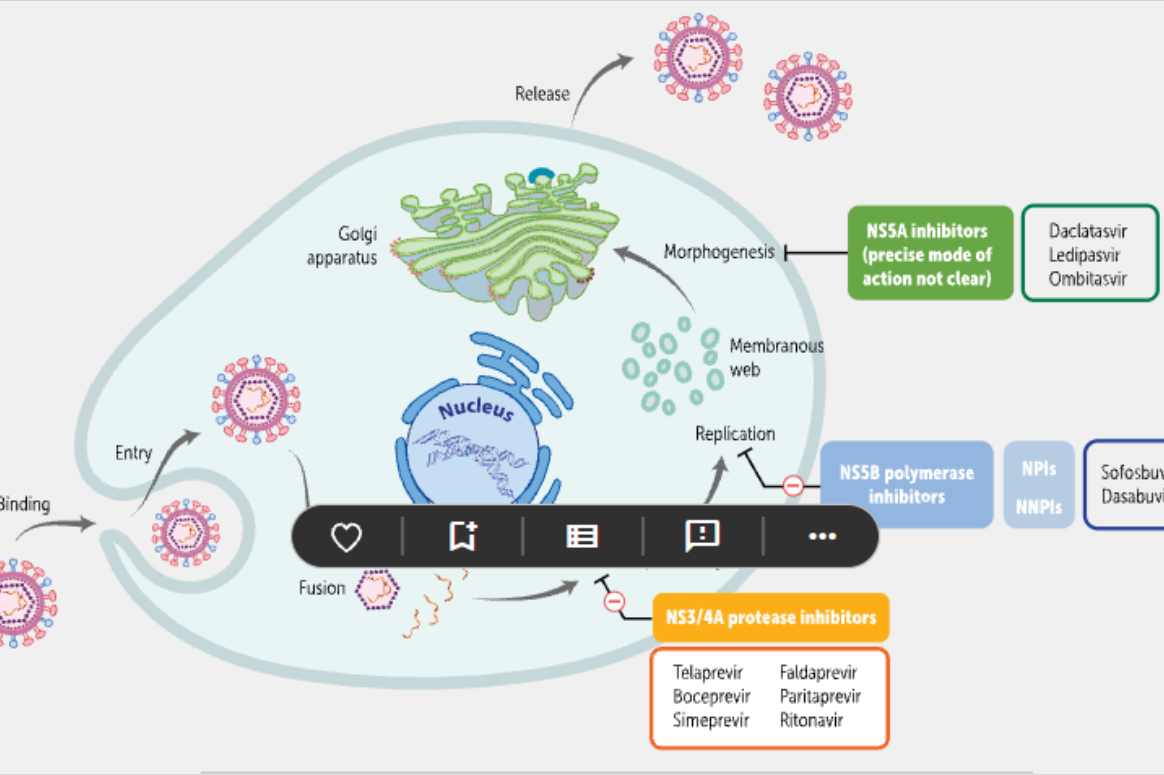
Most acute infections with hepatitis C virus (HCV) will progress to chronic HCV if left untreated. The goals of HCV treatment are viral eradication and prevention of complications (eg, cirrhosis, hepatocellular carcinoma).

Direct-Acting Antivirals (DAAs)

The new DAAs for chronic HCV have cure rates >95%. DAA regimens have better results, are better tolerated, and are not timing dependent. It is recommended that most patients with hepatitis C infection start treatment early to prevent the complications associated with long-standing disease.

DAAs are preferred because they have a high cure rate and fewer side effects. Interferon α-2a therapy has more side effects and a time limit for effectiveness.

Predetermined combinations of DAAs are used depending on treatment history, presence of cirrhosis, genotype, and the presence of resistance mutations. The goal of choosing a combination regimen is to effectively treat while minimizing drug interactions and drug toxicity. The main categories of treatment are NS3/4A protease inhibitors, NS5B polymerase inhibitors, and NS5A inhibitors, all unique to HCV (Figure 1).



QUIZ

Figure 3

NS3/4 protease inhibitors

These agents are inhibitors of NS3/4A serine protease which are involved in the post-translational replication of HCV.

Examples of NS3/4 protease inhibitors *(suffix as -previr)*: grazoprevir, voxilaprevir, glecaprevir.

Common adverse effects of these drugs include fatigue, headache, and nausea. These drugs are inhibitors of CYP3A4 and transporters such as P-gp (p-glycoprotein) and other transporters. They can cause significant drug interactions with NNRTs and PIs in patients with HIV/HCV co-infections.

NS5B RNA polymerase inhibitors

These drugs inhibit NS5B RNA polymerase. NS5B is an RNA-dependent RNA polymerase that can bind to nucleosides and non-nucleosides. Thus, inhibition of NS5B disrupts post-translational modifications and viral replication.

Examples of NS5B inhibitors *(suffix as -buvir)*: sofosbuvir.

Sofosbuvir is a P-gp substrate, and P-gp inducers should not be co-administered.

NS5A inhibitor

These agents inhibit NS5A protein and interfere with the viral assembly and replication. However, the exact mechanism is not clear.

Examples of NS5A inhibitors *(suffix -svir): elbasvir, ledipasvir, and pibrentasvir.*

Common adverse effects include headache, fatigue, and nausea. They are also inhibitors of the P-gp transporters and CYP1A2 and CYP3A4 enzymes.

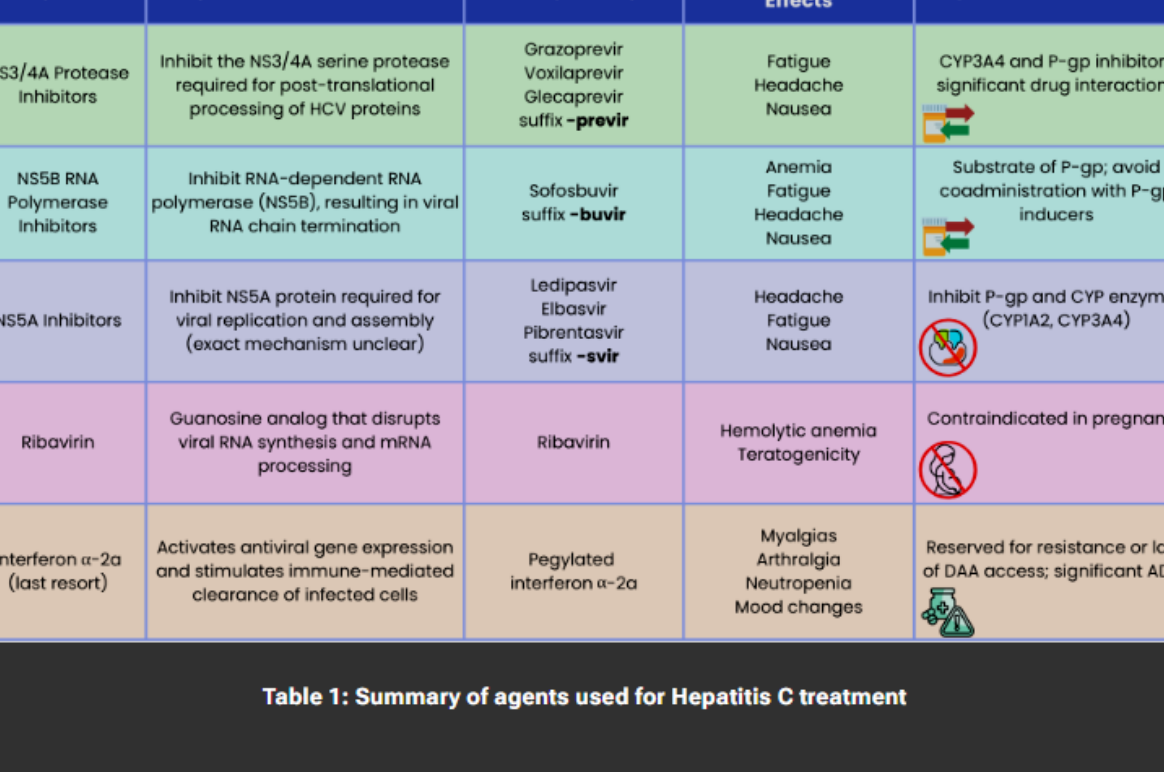
Ribavirin

Ribavirin is an antiviral drug that looks like guanosine, a normal building block of RNA. Ribavirin disrupts viral RNA production by lowering guanosine triphosphate levels, interfering with viral mRNA processing, and inhibiting viral RNA polymerase. As a result, it can slow the replication of many viruses, including influenza, respiratory syncytial virus, hepatitis C virus, and HIV.

Higher doses or longer courses of ribavirin may be more effective but increase the risk of side effects. The most important toxicity is dose-dependent hemolytic anemia, which occurs early in therapy.

Ribavirin is contraindicated in pregnancy (teratogenic) and two forms of contraception are required for both partners during treatment and for months afterward. It is contraindicated in patients with significant anemia, severe kidney disease, or COPD due to the risk of bronchoconstriction.

Similar to other antivirals, different drugs affect different stages of the viral life cycle (Table 1).



Drug Class	Target/Mechanism of Action	Example Drugs	Common Adverse Effects	Key Notes/Clinical Pearls
NS3/4A Protease Inhibitors	Inhibit the NS3/4A serine protease required for post-translational processing of HCV proteins	Grazoprevir Voxilaprevir Glecaprevir suffix **-previr**	Fatigue Headache Nausea	CYP3A4 and P-gp inhibitors → significant drug interactions
NS5B RNA Polymerase Inhibitors	Inhibit RNA-dependent RNA polymerase (NS5B), resulting in viral RNA chain termination	Sofosbuvir suffix **-buvir**	Anemia Fatigue Headache Nausea	Substrate of P-gp; avoid coadministration with P-gp inducers
NS5A Inhibitors	Inhibit NS5A protein required for viral replication and assembly (exact mechanism unclear)	Ledipasvir Elbasvir Pibrentasvir suffix **-svir**	Headache Fatigue Nausea	Inhibit P-gp and CYP enzymes (CYP1A2, CYP3A4)
Ribavirin	Guanosine analog that disrupts viral RNA synthesis and mRNA processing	Ribavirin	Hemolytic anemia Teratogenicity	Contraindicated in pregnancy
Interferon α-2a (last resort)	Activates antiviral gene expression and stimulates immune-mediated clearance of infected cells	Pegylated interferon α-2a	Myalgias Arthralgia Neutropenia Mood changes	Reserved for resistance or lack of DAA access; significant ADRs

Table 1: Summary of agents used for Hepatitis C treatment

Interferon α–2a as a Last Resort

Interferon α-2a treatment is only used if no other option is available, given viral resistance patterns or lack of insurance coverage of another drug. It consists of weekly injections for 24 weeks with genotype 1 infection and 12 weeks for other genotypes. Ribavirin can be added alongside the interferon for additional nucleoside inhibition.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AR ing a drug?

You explain to AR that modern hepatitis C treatment relies on direct-acting antivirals that are highly effective and well tolerated. Because her infection is chronic and she shows no evidence of cirrhosis, she is an excellent candidate for oral antiviral therapy aimed at viral eradication. You also clarify that certain drugs cannot be used. For instance, ribavirin is not an appropriate option for her treatment. Although ribavirin can interfere with viral RNA production, it is no longer a first-line agent for hepatitis C due to its significant toxicities, particularly hemolytic anemia, and its teratogenic potential, which makes it especially inappropriate for women of child-bearing age. Before finalizing a treatment regimen, you recommend that AR undergo HIV testing. You explain that hepatitis C and HIV can be transmitted through similar risk factors, including injection drug use, and that identifying HIV co-infection is important because certain antiviral medications can have clinically significant drug interactions with HIV therapies. Confirming her HIV status will ensure that the selected hepatitis C regimen is both safe and effective. You reassure AR that with appropriate evaluation and therapy, hepatitis C is now considered a curable disease, and early treatment dramatically reduces the risk of long-term complications such as cirrhosis and hepatocellular carcinoma.

Summary

- Hepatitis B and C are viral infections transmitted through different routes—sexual contact and blood exposure, respectively—and can lead to serious complications such as cirrhosis, liver cancer, and liver failure if left untreated.
- Chronic hepatitis B is treated with nucleoside and nucleotide analogs such as tenofovir and entecavir, which inhibit HBV DNA polymerase; treatment courses are long atient response.
- Hepatitis C is primarily treated with direct-acting antivirals (DAAs)—including NS3/4A protease inhibitors, NS5B polymerase inhibitors, and NS5A inhibitors—which achieve cure rates greater than 95% and are better tolerated than older therapies.
- Ribavirin, a guanosine analog that disrupts viral RNA production, may be used as an adjunct in hepatitis C treatment but carries significant risks including dose-dependent hemolytic anemia and is contraindicated in pregnancy and several comorbid conditions.
- Interferon α-2a can be used as a last resort for both hepatitis B and C when other options are unavailable, but is associated with a higher frequency of adverse effects—including myalgias, neutropenia, and mood disturbances—and requires close patient monitoring.

Review Questions

1. A patient with a history of intravenous drug use presents with chronic right upper quadrant pain and elevated transaminases. Viral studies demonstrate HBV/HCV co-infection. Which of the following medications can be used to address both viral infections?

A. ○ Adefovir
B. ○ Dasabuvir
C. ○ Interferon α-2a
D. ○ Lamivudine
E. ○ Sofosbuvir

Explanation (requires correct answer)

2. One week after beginning treatment for a viral hepatitis, your patient comes in with "heavy legs." On exam, you notice bilateral 2+ pitting edema of the lower extremities. Which of the following medications was most likely started?

A. ○ Daclatasvir
B. ○ Entecavir
C. ○ Lamivudine
D. ○ Ribavirin
E. ○ Telbivudine

Explanation (requires correct answer)

3. A patient with chronic HCV is started on a new medication. Two weeks later, labs show significant anemia. Which medication was most likely started?

A. ○ Sofosbuvir
B. ○ Elbasvir
C. ○ Ribavirin
D. ○ Glecaprevir

Explanation (requires correct answer)

4. A 55-year-old man with chronic hepatitis C is started on a direct-acting antiviral regimen. The prescribed drug functions as a nucleoside analog that inhibits the viral RNA–dependent RNA polymerase, resulting in termination of viral RNA synthesis. Which of the following agents was most likely prescribed?

A. ○ Ledipasvir
B. ○ Entecavir
C. ○ Sofosbuvir
D. ○ Ribavirin

Explanation (requires correct answer)

D. ○ Ribavirin

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ Compare and contrast the mechanisms of action of agents used in the treatment of hepatitis B infections.
02 ☐ Compare and contrast agents the mechanism of action of agents used in the treatment of hepatitis C infections.
03 ☐ Discuss the adverse drug reactions of drugs used to treat hepatitis B and C infections.

Objective Confidence: 0/3